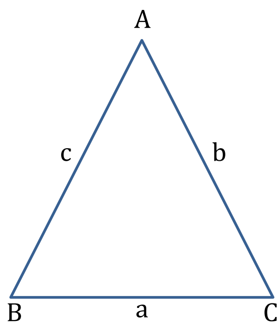
6 Heron's Formula

In a scalene triangle, if the length of each side is given but its height is not known and it cannot be obtained easily, we take the help of Heron's formula or Hero's formula given by Heron to find the area of such a triangle.

Heron's formula : If a, b, c denote the lengths of the sides of a triangle ABC. Then,



$$\text{Area of } \Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$$

where s is the semi-perimeter of ΔABC.

$$s = \frac{a+b+c}{2}$$

Quick Tips

✸ This formula is applicable to all types of triangles whether it is right-angled or equilateral or isosceles and it is also useful in finding the area of a triangle when it is not possible to find the area of the triangle easily.

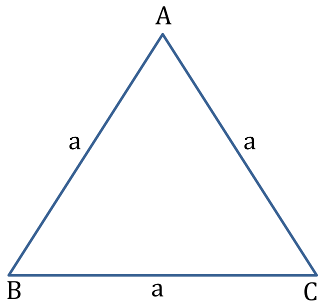
Do You Remember ?

- ✸ In earlier classes you have learned about Heron's formula to find the area of triangle when all 3 sides are known to us.

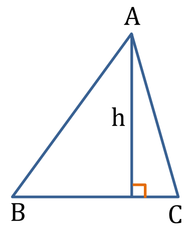
  It this chapter, we will continue to use Heron's formula to find area of Δs and even extend its use for finding the area of quadrilateral.

Important formulas:

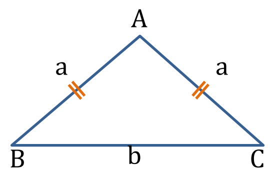
(1) Area of equilateral triangle = $\frac{\sqrt{3}}{4} \times \text{Side}^2 = \frac{\sqrt{3}}{4}a^2$



(2) Area of scalene triangle = $\frac{1}{2} \times \text{Base} \times \text{Height}$



(3) Area of isosceles triangle = $\frac{1}{2} \times \sqrt{\left(a^2 - \frac{b^2}{4}\right)} \times b$



Numerical Ability 1

Find the area of a triangle whose sides are 13 cm, 14 cm and 15 cm respectively.

Solution

Let a, b, c be the sides of the given triangle and s be its semi-perimeter such that a = 13 cm, b = 14 cm and c = 15 cm

Now, $s = \frac{(a + b + c)}{2} = \frac{(13 + 14 + 15)}{2} = 21$

$\therefore$ $s - a = 21 - 13 = 8, s - b = 21 - 14 = 7$ and $s - c = 21 - 15 = 6$

Hence, Area of given triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{21 \times 8 \times 7 \times 6}$

$= \sqrt{7 \times 3 \times 2 \times 2 \times 2 \times 7 \times 2 \times 3}$

$= 7 \times 4 \times 3 = 84 \text{ cm}^2$

Building Concepts 1

The perimeter of a triangular field is 450 m and its sides are in the ratio 13 : 12 : 5. Find the area of triangle.

Explanation

It is given that the sides a, b, c of the triangle are in the ratio 13 : 12 : 5

i.e., $a : b : c = 13 : 12 : 5$

$\Rightarrow$ $a = 13x, b = 12x$ and $c = 5x$

$\therefore$ Perimeter = 450

$\Rightarrow$ $13x + 12x + 5x = 450$

$\Rightarrow$ $30x = 450$

$\Rightarrow$ $x = 15$ m

So, the sides of the triangle are

$a = 13 \times 15 = 195$ m, $b = 12 \times 15 = 180$ m

and $c = 5 \times 15 = 75$ m

It is given that perimeter = 450

$\Rightarrow$ $2s = 450$

$\Rightarrow$ $s = 225$ m

Hence, Area = $\sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{225(225-195)(225-180)(225-75)}$

$\Rightarrow$ Area = $\sqrt{225 \times 30 \times 45 \times 150}$

$= \sqrt{5^2 \times 3^2 \times 3 \times 5 \times 2 \times 3^2 \times 5 \times 5^2 \times 2 \times 3}$

$\Rightarrow$ Area = $\sqrt{5^6 \times 3^6 \times 2^2}$

$= 5^3 \times 3^3 \times 2 = 6750 \text{ m}^2$

SPOT LIGHT

In geometry, Heron's (or Hero's) formula, is named after Heron of Alexandria an ancient Greek mathematician and engineer who was active in his native city of Alexandria, Roman Egypt.

Do You Remember ?

★ Squares of 25 natural numbers are $1^2 = 1, 2^2 = 4, 3^2 = 9, 4^2 = 16$ ……, $25^2 = 625$

Numerical Ability 2

Find the area of a triangle having perimeter 32 cm, one side 11 cm and difference of other two sides is 5 cm.

Solution

Let a, b and c be the three sides of ΔABC.

$a = 11$ cm

$a + b + c = 32$ cm

$\Rightarrow \quad 11 + b + c = 32$ cm

or $\quad b + c = 21$ cm ... (i)

Also, we are given that $b - c = 5$ cm ... (ii)

Adding (i) and (ii),

$2b = 26$ cm

i.e., $\quad b = 13$ cm and $c = 8$ cm

Now, $s = \dfrac{a+b+c}{2}$

$= \dfrac{11+13+8}{2} = \dfrac{32}{2} = 16$ cm

$(s - a) = (16 - 11)$ cm $= 5$ cm

$(s - b) = (16 - 13)$ cm $= 3$ cm

$(s - c) = (16 - 8)$ cm $= 8$ cm

$\therefore \quad$ Area of $\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{16\times5\times3\times8}$ cm^2

$= \sqrt{64\times30}$ cm^2

$= 8\sqrt{30}$ cm^2

SPOT LIGHT

Area of a right angled isosceles triangle

Let length of the equal sides of triangle be 'a' and $\angle A = 90°$ then

(i) Area $= a^2/2$

(ii) Perimeter $= 2a + \sqrt{2}a$

Numerical Ability 3

The length of the sides of a triangle are 5 cm, 12 cm and 13 cm respectively. Find the length of perpendicular from the opposite vertex to the side whose length is 13 cm.

Solution

Here, $a = 5$ cm, $b = 12$ cm and $c = 13$ cm

$\therefore \quad s = \dfrac{1}{2}(a + b + c)$

$= \dfrac{1}{2}(5 + 12 + 13)$

$= \dfrac{30}{2} = 15$ cm

Let A be the area of the given triangle. Then,

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{15(15-5)(15-12)(15-13)}$$

$$\Rightarrow \quad A = \sqrt{15\times10\times3\times2} = 30 \text{ cm}^2 \qquad \text{... (i)}$$

Let p be the length of the perpendicular from vertex A to the side BC. Then,

$$A = \frac{1}{2} \times 13 \times p \qquad \text{... (ii)}$$

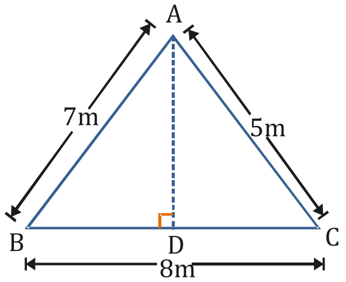
From (i) and (ii), we get

$$\Rightarrow \quad \frac{1}{2} \times 13 \times p = 30$$

$$\Rightarrow \quad p = \frac{60}{13} \text{ cm}$$

Numerical Ability 4

In the figure, there is a triangular children park with sides, AB = 7 m, BC = 8 m and AC = 5m, AD ⊥ BC and AD meets BC at D. Trees are planted at A, B, C and D. Find the distance between the trees at A and D.



Solution

In figure, a = 8m, b = 5m and c = 7m

$$s = \frac{8+5+7}{2} \text{ m} = \frac{20}{2} = 10\text{m}$$

The area of $\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$

$$= \sqrt{10\times(10-8)\times(10-5)\times(10-7)} \text{ m}^2$$

$$= \sqrt{10\times2\times5\times3} \text{ m}^2 = 10\sqrt{3} \text{ m}^2$$

Now, AD is perpendicular to BC.

$$\Rightarrow \quad \frac{1}{2} \times BC \times AD = 10\sqrt{3}$$

$$\Rightarrow \quad \frac{1}{2} \times 8 \times AD = 10\sqrt{3}$$

$$\Rightarrow \quad AD = \frac{10\sqrt{3}}{4} \text{ m} = \frac{5\sqrt{3}}{2} \text{ m}$$

Hence, the distance between the trees at A and D is $\frac{5\sqrt{3}}{2}$ m.

In the golden triangle, its three angles are in 2 : 2 : 1 proportion. [Base angles are 2θ, 2θ and vertical angle is θ]

SPOT LIGHT

Quick Tips

- ✸ Sometimes it is not required to use Heron's formula as the length of sides forms "Pythagorean Triplet".

  Examples of Pythagorean Triplet are:

  3 units, 4 units, 5 units; 6 units, 8 units, 10 units; 12 units, 5 units, 13 units etc.

- ✸ You might have observed that while solving the questions related to Heron's formula, we get the answers as irrational numbers (having radical sign).

  In most of the questions the values of these irrational terms are given but if the value is not given than we can follow the method of estimation to find the approx. answer.

E.g. Let us try to find the value of $\sqrt{10}$ by estimation.

By inspection we can see

$3^2 < 10 < 4^2$

$3 < \sqrt{10} < 4$

By hit and trial, we can find that value of $\sqrt{10}$ will be between 3.1 and 3.2 as

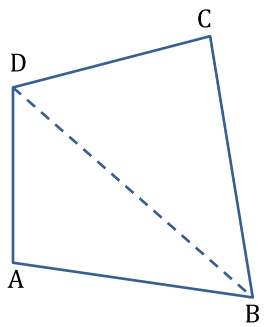
$(3.1)^2 = 9.61$ and $(3.2)^2 = 10.24$

But 10 is more closer to $(3.2)^2 = 10.24$ so we can take estimated value of $\sqrt{10}$ as 3.2.

- ✸ To find estimated roots quickly and accurately, it is advisable to learn the squares of first 25 natural numbers.

Applications of Heron's formula in finding area of a quadrilateral

Heron's formula can be applied to find the area of a quadrilateral by dividing the quadrilateral into two triangular parts. If we join any of the two diagonals of the quadrilateral, then we get two triangles. Area of each triangle is calculated and the sum of two areas is the area of the quadrilateral.

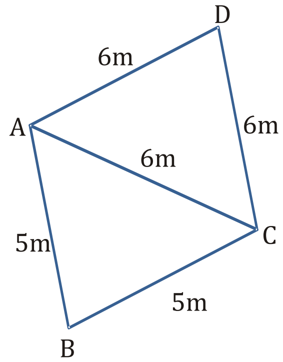


Area of quadrilateral ABCD = Area of ΔABD + Area of ΔBCD

Numerical Ability 5

Prove that the area of the quadrilateral ABCD is $3(4+3\sqrt{3})m^2$, If AB = 5 m, BC = 5 m, CD = 6 m, AD = 6 m, and diagonal AC = 6 m.

Solution



Diagonal AC divides the quadrilateral ABCD into two triangles ΔACD and ΔABC.

For ΔACD, sides are 6 m, 6 m and 6 m.

semi-perimeter, s = $\dfrac{6m+6m+6m}{2}$ = 9 m

$\therefore$ Area of ΔACD = $\sqrt{s(s-a)(s-b)(s-c)}$

$=\sqrt{9\times(9-6)(9-6)\times(9-6)}$ m²

$=\sqrt{9\times3\times3\times3} = 9\sqrt{3}$ m²

For ΔABC, sides are 5 m, 5 m and 6 m.

semi-perimeter, s = $\dfrac{5m+5m+6m}{2}$ = 8m

Area of ΔABC = $\sqrt{s(s-a)(s-b)(s-c)}$

$=\sqrt{8(8-5)(8-5)(8-6)}$

$=\sqrt{8\times3\times3\times2}$

$=\sqrt{16\times9}$ m² = 12 m²

Thus, the area of the quadrilateral ABCD = $(12 + 9\sqrt{3})$m² = $3(4 + 3\sqrt{3})$m²

Hence proved.

Building Concepts 2

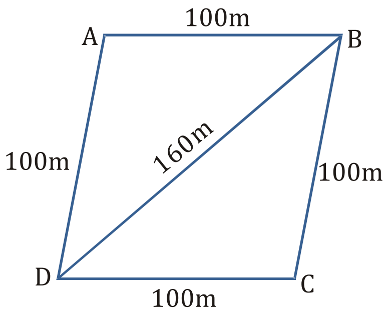
Sanya has a piece of land which is in the shape of a rhombus. She wants her one daughter and one son to work on the land and produce different crops to suffice the needs of their family. She divided the land in two equal parts. If the perimeter of the land is 400 m and one of the diagonals is 160 m, how much area each of them will get for their crops?

Explanation

Let ABCD be the field which is divided by the diagonal BD = 160 m into two equal parts.

Since ABCD is a rhombus of perimeter 400 m. Therefore,

$AB = BC = CD = DA = \frac{400}{4} m = 100 m$



Now, consider ΔBCD

Let s be the semi-perimeter of ΔBCD.

Then, $s = \frac{BC + CD + BD}{2}$

$= \frac{100 + 100 + 160}{2} m$

$= 180 m$

So, area of $\Delta BCD = \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{180(180-100)(180-100)(180-160)}$

$= \sqrt{180 \times 80 \times 80 \times 20} = 4800 m^2$

Since, diagonal of rhombus divides it into two equal areas.

Hence each of two children will get an area of 4800 m^2.

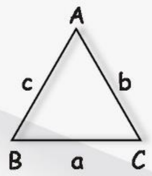
Memory map

Heron's Formula

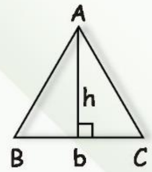
Scalene triangle

Area of $\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$

$S = \dfrac{a+b+c}{2}$

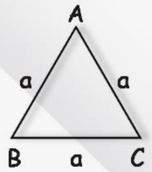


Area of triangle



$= \dfrac{1}{2} \times \text{base} \times \text{height}$

Equilateral triangles



$\text{Area} = \dfrac{\sqrt{3}}{4} \times (a)^2$

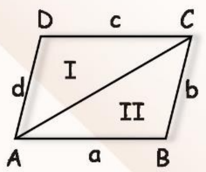
$a \rightarrow$ Side

Units of measurement

Length & Breadth $\rightarrow$ m/cm

Area = m^2/cm^2

Area of quadrilateral



Whose one side & one diagonal are given, can be calculated by dividing the quadrilateral into two triangles and using Heron's formula.